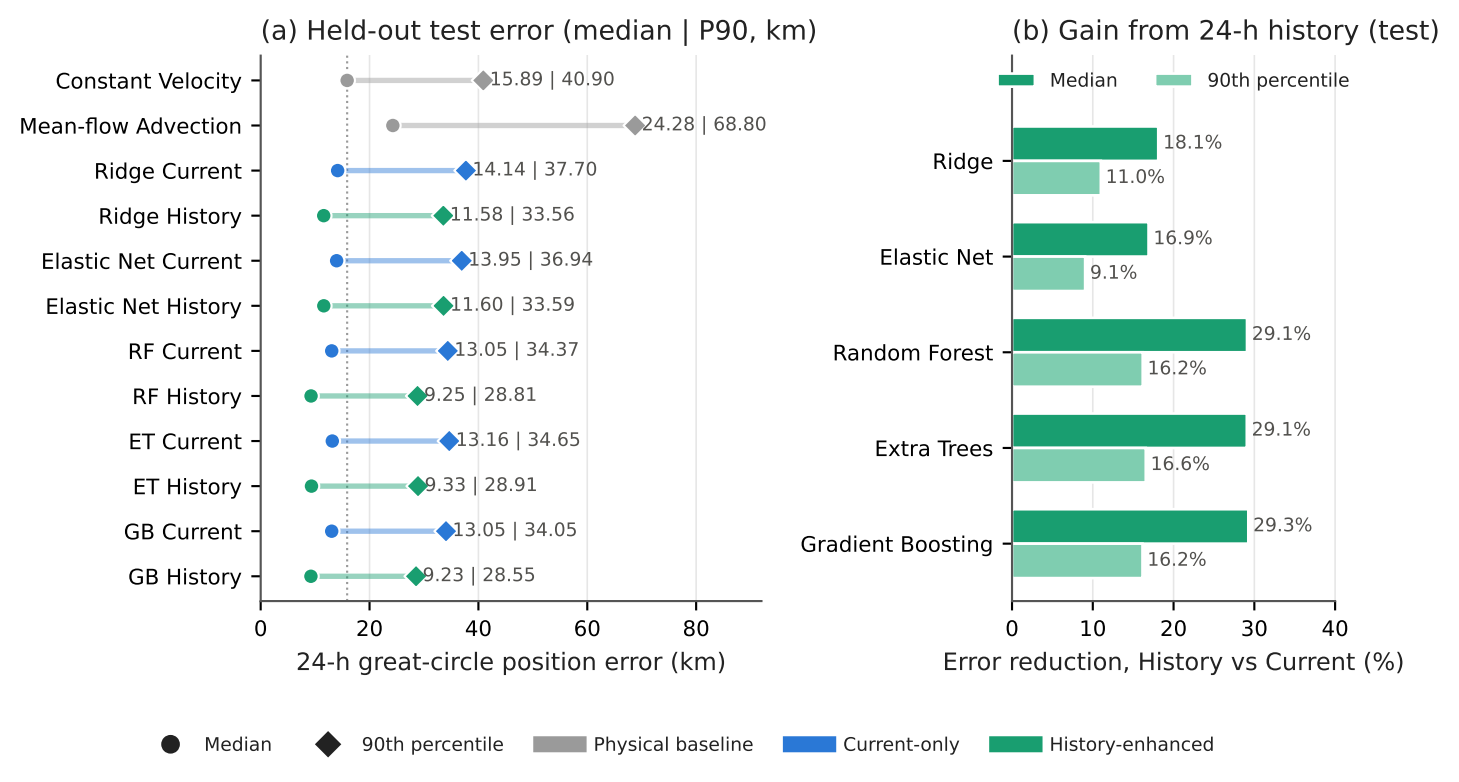
(a) Held-out test error (median | P90, km)
Constant Velocity
Mean-flow Advection
Ridge Current
Ridge History
Elastic Net Current
Elastic Net History
RF Current
RF History
ET Current
ET History
GB Current
GB History
15.89 | 40.90
24.28 | 68.80
14.14 | 37.70
11.58 | 33.56
13.95 | 36.94
11.60 | 33.59
13.05 | 34.37
9.25 | 28.81
13.16 | 34.65
9.33 | 28.91
13.05 | 34.05
9.23 | 28.55
0
20
40
60
80
24-h great-circle position error (km)
(b) Gain from 24-h history (test)
Median
90th percentile
Ridge
18.1%
11.0%
Elastic Net
16.9%
9.1%
Random Forest
29.1%
16.2%
Extra Trees
29.1%
16.6%
Gradient Boosting
29.3%
16.2%
0
10
20
30
40
Error reduction, History vs Current (%)
Median
90th percentile
Physical baseline
Current-only
History-enhanced